



ONE SHOT

REVISION SERIES

CLASS 10 **GEOGRAPHY**

MINERAL AND

ENERGY RESOURCES



Introduction

What are minerals?



Minerals are an indispensable part of our lives. [Pin to ship]



"Homogenous naturally occurring substance with a definable internal structure"

Rocks Minerals



Rocks are combinations of homogenous substances called mineral [One or more than one]



Formation depends upon the physical and chemical conditions. 🤔

Geographers

- Study mineral as a part of earth crust.
- Above the earth.

v/s

Geologists

- Study formation of minerals, their age physical and chemical composition.
- Below the earth.

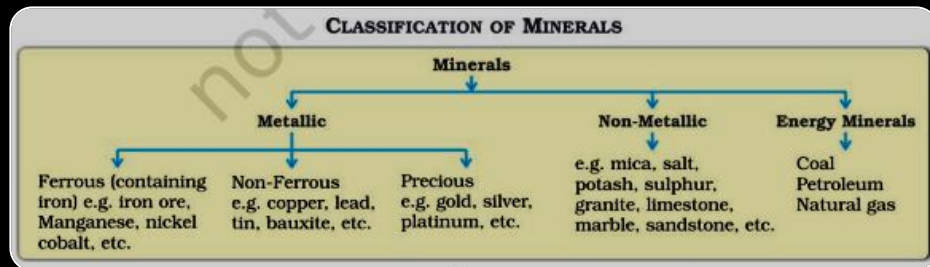
Mode of Occurrence of Minerals

Found in “Ores” [Minerals mixed with other element]

1. **Veins and lodes** - E.g. Tin, copper and Zinc.
2. **Beds and layers** - E.g. Coal and gypsum.
3. **Residual mass of weathered material** - E.g. Bauxite
4. **Alluvial deposits** - E.g. Gold, silver and tin.
5. **Ocean water** - E.g. Salt, magnesium and bromine.

Distribution of minerals

- **Peninsular rocks**
 - Coal, metallic minerals, mica, and many other non-metallic minerals.
- **East and West of peninsula**
 - Petroleum
- **Rajasthan**
 - Non-ferrous minerals [Copper]
- **North India**
 - Devoid of economic minerals.



Ferrous Minerals

$\frac{3}{4}$ of total metallic minerals, base for metallurgical industries.

Iron ore

- Backbone of industrial development.
- **Magnetite** [70% of iron content].
- Excellent magnetic qualities.
- **Hematite** [50-60% of content].
- Industrial iron.

Manganese

- Used in making steel ferro – manganese alloy.
- Used in manufacturing bleaching powder, insecticides, and paints.

In 2018–19 almost entire production of iron ore (97%) accrued from Odisha, Chhattisgarh, Karnataka and Jharkhand. The remaining production (3%) was from other states.

Minerals and Energy Resources

Major iron ore belts in India

Orissa - Jharkhand belt

- Hematite are found in Badampahar, Gua and Noamundi.
- Port → Paradwip port.

Durg - Bastar - Chandrapur

- Chhattisgarh and Maharashtra.
- Hematite are found in Bailadila range of Bastar district.
- Port → Visakhapatnam port.

Bellary - Chitradurga - Chikmagalur - Tumkur

- Kudremukh mines is 100% export unit.
- Port → Mangalore [Through pipeline]

Maharashtra - Goa belt

- Goa and Ratnagiri district of Maharashtra
- Port → Marmagao.

Non Ferrous Minerals

- Not sufficient
- Mineral such as **copper, Bauxite, Zinc, Lead and Gold.**
- Used in metallurgical engineering and electrical.



Copper

- India is critically deficient in copper production.
- Malleable, Ductile and the good conductor. Therefore have high demand.
- The Balaghat mines in Madhya Pradesh, Khetri mines in Rajasthan and Singhbhum district of Jharkhand are leading producers of copper.

Bauxite

- Bauxite → **Alumina** → **Aluminium**
- It combines strength of metal **E.g.** Iron with extreme lightness, good conductivity and great malleability.
- Mainly found in Amarkantak plateau, Maikal hills plateau region of Bilaspur-Katni.
- **Orissa** → largest producing bauxite state.

Non-Metallic Minerals and Rock Minerals

Mica

- Modes of series of plates.
- Used in electronic industries due to dielectric strength, low power loss factor, insulating properties and resistance to high voltage.
- Chota Nagpur plateau, Koderma
- Gaya-Hazaribagh, Ajmer, Rajasthan and Nellore Mica belt of A.P.



Limestone

- Rock composed of calcium carbonates or calcium and magnesium carbonate.
- Essential for smelting iron ore in Blast furnace and cement industries.
- A.P., M.P., Rajasthan, Gujarat, T.N. and many more.



Hazards of Mining and Need for Conservation

Impact on miners

↓
Health issues

- Dust and noxious fumes and inhaled.
- Vulnerable to pulmonary disease.

Impact on Environment

Water source get
contaminated

Dumping of waste
and slurry

↓
Degradation of land, soil
and river pollution.

Conservation of minerals

- Our dependency on minerals and availability to us.
- 1% of earth's crust.
- Formation $\neq$ Consumption.

∴ **Finite and non-renewable.**

- Continued extraction → Cost [Greater depth, decreased quality]

Steps for conservation

- Improved technology, recycling of metals, using other alternatives and substitutes.

Energy Resources



Need 🤔

Conventional Source

- Coal
- Petroleum
- Natural gas
- Electricity

Non Conventional Source

- Nuclear or Atomic energy
- Solar energy
- Wind energy
- Bio-gas
- Tidal energy
- Geothermal energy.

Coal

Coal is an extremely important source of energy. It is used for power generation, to supply energy to industry as well as for domestic needs.



Formation depends upon degree of compression, depth and time of burial.

Peat

- Low carbon.
- High moisture.
- ∴ Low heating capacity.

Lignite

- Low grade brown coal.
- Soft with high moisture content.

Bituminous

- Deep inside the earth.
- Temperature.
- ∴ Commercial use, smelting.

Anthracite

- Highest quality
- Hard coal

Gondwana Coal**Tertiary Coal**

On the basis of age →

- 200 million ago.
- Damodar valley, Jharia, Raniganj, Bokaro.

- 55 million ago.
- North eastern state.

Remember coal is a bulky material, which loses weight on use as it is reduced to ash. Hence, heavy industries and thermal power stations are located on or near the coalfields.

Petroleum and Natural Gas

Petroleum

- Petroleum industry as a **“Nodal Industry”**. 🧐
- Anticlines and fault traps in rocks formed during tertiary age.
- Porous and non porous layer and gas been lighter usually occurs above the oil.
- Found in - Mumbai high, Gujarat [Ankleshwar], Assam [Digboi].

Natural Gas

- Source of energy as well as an industrial raw material.
- Found in association with or without petroleum.
- Environmental friendly Low carbon dioxide emissions.
- Krishna, Godavari, Basin, Mumbai High, Gulf of Cambay.
- **HVJ pipeline** [Hazira-Vijaipur-Jagdishpur] = Artery to Indian gas production.
- Power and fertilizer industries are key users.
- CNG [Compressed Natural Gas] is used in vehicles is replacing liquid fuels.

Electricity

- It's Importance. 🤔
- Per capita consumption is considered as index of development.

Hydro Electricity

- Produced by running water.
- Use renewable source.
- Multipurpose river projects like Bhakra nangal, Damodar valley crop.

Thermal Electricity

- By burning coal petroleum and natural gas.
- Use non renewable fossil fuel.

Non-Conventional Source of Energy

Need for it



Consumption of fossil fuels



Depletion of resources



 Price rise



Uncertainties in future and environmental problems.



We must use renewable source of energy.

E.g. - Solar energy, wind energy, tide and biomass

Nuclear or Atomic Energy

- Obtained by altering the structure of atoms.
- Uranium and thorium are used.
- Found in Jharkhand, the Aravalli ranges of Rajasthan and the monazite sand of Kerala. [rich in thorium]

Solar Energy

- Photovoltaic technology converts sunlight directly into electricity.
- Solar energy is fast becoming popular in rural and remote areas.
- Solar plants will contribute to environmental conservation and adequate supply of manure in agriculture.

Non-Conventional Source of Energy**Wind Energy**

- Potential of wind.
- Largest wind farm cluster Nagercoil to Madurai [T.N.]
- A.P., Karnataka, Gujarat, Kerala, Maharashtra, etc

Bio Gas

- Shrubs, farm waste, animal and human waste is used to make biogas.
- “Gobar Gas plant” → **Twin benefits.**

Source of energy

Quality of manure.

+

Prevents the loss of trees.

Tidal Energy

- Oceanic tides can be used to generate electricity.
- The Gulf of Khambhat, the Gulf of Kachchh in Gujarat on the western coast and Gangetic delta in Sunderban regions of West Bengal provide ideal conditions for utilising tidal energy.

Geothermal Energy

- Refers to the heat and electricity produced by using the heat from the interior of the Earth.
- The Parvati valley near Manikaran in HP and Puga valley Ladakh.

Conservation of Energy Resources

Increasing development

=

Need for the energy increase.

∴ Consumption of energy is also increasing.



We should use sustainable energy.

- Use public transportation.
- Switch off electricity when not in use.
- Using power saving devices.
- Using non-conventional sources of energy.

Energy saved is energy produced.



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